Joseph Amal A.

Aspiring AI / ML Engineer · M.Sc. Software Systems

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Summary

M.Sc. Software Systems student building towards a career as an AI / ML engineer, who builds AI systems end to end and ships them as working apps: deep learning for audio, RAG with cited answers, and LLM-powered coaching. Open to **AI Engineer**, **ML Engineer** and **Software Developer** roles.

Experience

Nanonino Jul 2026 – Oct 2026
Intern (three-month internship) Coimbatore

- Built the **Synthetic Voice Detection System** end to end as the sole developer (see Projects).
- Built **CaddieAI** for the courserev.ai team: a routed RAG assistant with reranking and cited answers; **111/112** questions routed correctly and **25/25** hard questions answered.
- Built the **RAG Knowledge Base API** with a fellow intern: NestJS, hybrid search in Upstash Vector, served over REST and as an MCP tool for a voice bot.

Projects

Synthetic Voice Detection System / *Python, PyTorch, Librosa, Streamlit* Aug 2026 – Sep 2026
Deep learning · voice anti-spoofing · internship project at Nanonino Live demo · GitHub

- Detects AI-generated and cloned voices: MFCC features (40 × 400) from each clip go through a **CNN + BiLSTM** that returns Genuine or Synthetic with a confidence score.
- Fixed a domain mismatch with joint training on ASVspoof 2019 and In-the-Wild data plus weighted sampling: real-world accuracy **46.8% → 96.1%**, **EER 3.60%**, while keeping 97.6% on studio audio.
- Catches **92–100%** of neural text-to-speech attacks (ASVspoof eval, 71,237 clips); a 0.8 MB model served in a live web app with recording, an adjustable threshold and spectrogram / MFCC evidence.

Confident Talk Analyzer / *Python, Flask, Whisper, wav2vec2, LLM, PostgreSQL, Docker* Dec 2025 – Sep 2026
Speech-coaching web app · M.Sc. mini project, rebuilt Live demo · GitHub

- Records an answer in one of **8 practice modes** (interview, JAM, presentation, pitch...), transcribes it with Whisper and scores **10 delivery skills**, including wav2vec2 vocal confidence.
- An LLM coach reviews the answer against frameworks such as STAR, with guardrails that removed invented numbers from its tips (6 of 12 → none).
- Raised hesitation-pause detection F1 **33% → 95%**; **265 automated tests** on SQLite and PostgreSQL in CI; deployed on Modal.

Other projects: CaddieAI (routed RAG golf assistant) · RAG Knowledge Base API (NestJS, MCP) · Automatic Garage Design using Blynk Cloud Platform (IoT, ICAC-2024 paper) · Blockchain Technology in Decentralised Finance (ATNEAI 2026 paper)

Education

Sri Krishna Arts and Science College (Bharathiar University) Coimbatore
Master of Software Systems, 5-year integrated programme · CGPA 7.9 / 10 2023 – present

Trinity Matriculation Higher Secondary School 83%
Higher Secondary Certificate (HSC) 2021 – 2023

Technical Skills

Languages:	Python, Java, C++, SQL, TypeScript, JavaScript
ML / Deep Learning:	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, CNN-LSTM, fine-tuning
LLMs & GenAI:	RAG, LangChain, embeddings, vector databases (Pinecone, Upstash), reranking, LLM evaluation, prompt engineering, MCP
Speech & Audio:	Whisper, wav2vec2, Librosa, MFCC feature extraction
Backend & MLOps:	FastAPI, Flask, NestJS, PostgreSQL, Docker, GitHub Actions, pytest, ONNX Runtime, Streamlit, Git

Achievements & Certifications

- Papers presented at **ICAC-2024** (International Conference on Advanced Computing, Bharathiar University) and **ATNEAI 2026** (13th National Conference).
- **Smart India Hackathon 2025**: proposed TripZap (team Optiminds), tourist safety with AI, geo-fencing and a blockchain-based digital ID.
- **First Prize, Best Entrepreneur** ("Foundera"); recognised for academic excellence; Event Coordinator of the college department.
- **Certifications:** Generative AI & prompt engineering (9, Vanderbilt University via Coursera); Python (2, University of Michigan & Coursera).
- **Languages:** English, Tamil (fluent); Malayalam (intermediate); French (basic).